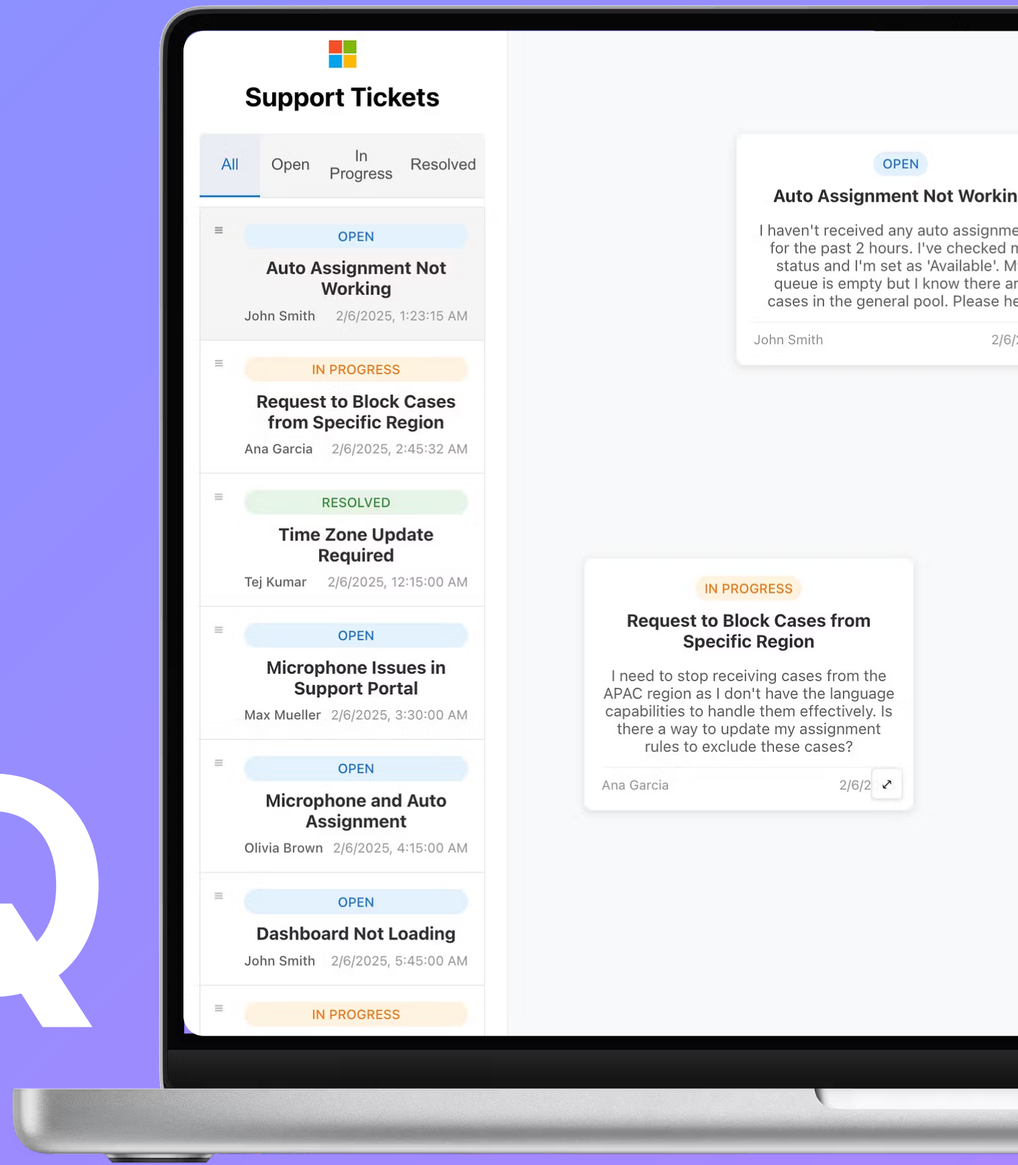
RESOLVEIQ

A MICROSOFT SPONSORED CAPSTONE PROJECT



Try Pitch

Meet Our Team



NOOR RAJAN

PRODUCT MANAGER
UI/UX RESEARCHER



KUNAL SRIVASTAVA

SOFTWARE ENGINEER



CONNOR CHAN

UI/UX DESIGNER



AYAAN RAHIM

SOFTWARE ENGINEER



NATHAN LEE

SOFTWARE ENGINEER

Try Pitch

PROBLEM CONTEXT

Microsoft's current internal operations support for Intelligent Assignment relies on manual processes and does not leverage Generative AI capabilities to improve resolution times

Try Pitch

Key Pain Points

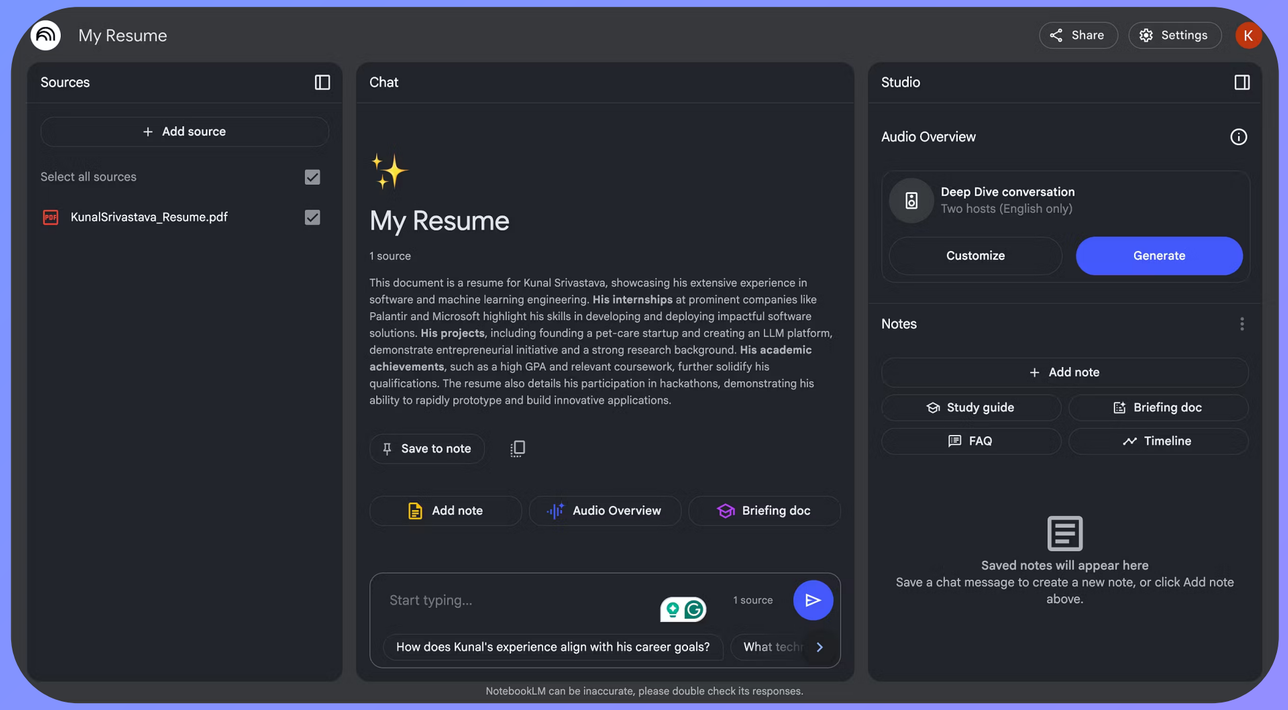
1. Slow resolution times due to manual search
2. High chance of missing critical information
3. Inconsistent responses impacting customer satisfaction
4. No automated feedback loop to improve future resolution efficiency

Try Pitch

How might internal operation teams achieve more **efficient** and **accurate** resolving of customer support tickets so that they can **reduce** response times and **improve** resolution rates?

Try Pitch

Key Research Insights



Screenshot of NotebookLM (from Google)

67%

OF ORGANIZATIONS HAVE ALREADY BEGUN USING GENERATIVE AI IN CUSTOMER SERVICE FUNCTIONS

IBM Research

#1

METRIC SUPPORT TEAMS TRACK FOR MEAN TIME-TO-RESOLUTION (MTTR)

AtomicWork Research

MICROSOFT INTERNAL OPERATIONS SUPPORT

20,000

INTERNAL USERS GLOBALLY

4,000

REQUESTS

Microsoft

INEFFECTIVE KNOWLEDGE MANAGEMENT SIGNIFICANTLY IMPACTS SUPPORT PERFORMANCE METRICS

Knoweledge AI

Try Pitch



MICHAEL K. |
OPERATIONS/HELPDESK AGENT

AGE 40, SEATTLE, WA
MICROSOFT

Persona 1

Goals:

- Resolve support tickets quickly while minimizing back-and-forth communication
- Craft clear and concise responses without wasting time searching for missing details
- Identify execution steps to resolve issues in knowledge base

Pain Points:

- Spends too much time searching for missing ticket details
- Lacks intuitive ways to categorize and manage tickets
- Deals with inefficiencies across multiple platforms

Needs:

- Additional details readily available on support tickets
- More intuitive ticket categorization and management system
- Consolidated platform that reduces platform-switching
- Tools that streamline response and ticket resolution

Desired Outcomes:

- More efficient ticket resolution process
- One platform

Try Pitch



ARIA S. | MANAGER/LEAD

AGE 30, NEW YORK, NY
MICROSOFT

Persona 2

Goals:

- Optimize her team's support workflow and ensure efficient ticket resolution
- Establish a system that provides clear ticket statuses and workload distribution

Pain Points:

- Struggles with inefficiencies in tracking support tickets
- Faces challenges in managing escalations effectively

Needs:

- Enhanced support operations management capabilities

Desired Outcomes:

- Improved ticket resolution efficiency
- Better workload distribution across team members
- Proactive issue identification and resolution
- Enhanced overall support operations performance

Try Pitch

KEY CONCEPTS

1

Automation

UTILIZE RAG TO PARSE THROUGH KNOWLEDGE WIKI AND CRM

2

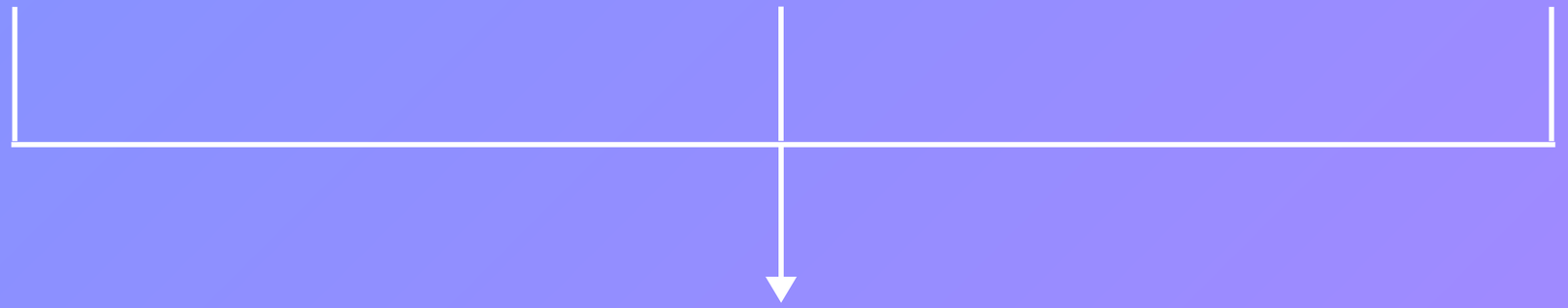
Reduce resolution times

UTILIZE LLM TO GENERATE A SUMMARIZED SOLUTION

3

Accurate responses

FEEDBACK LOOP TO IMPROVE SYSTEM ACCURACY



✓ IMPROVE CUSTOMER SATISFACTION

Try Pitch

Concept Validation

01

USERS

PRIMARY USERS: MICROSOFT SUPPORT AGENTS AND MANAGERS

02

RESEARCH

CONDUCTED USER INTERVIEWS & WEEKLY STANDUPS FOR FEEDBACK USABILITY TESTS WITH ENGINEERS & ONGOING ITERATIVE TESTING

03

SOLUTION

COMBINED KNOWLEDGE WIKI, RESPONSE CAPABILITY, & RECOMMENDED STEPS INTO ONE PLATFORMS

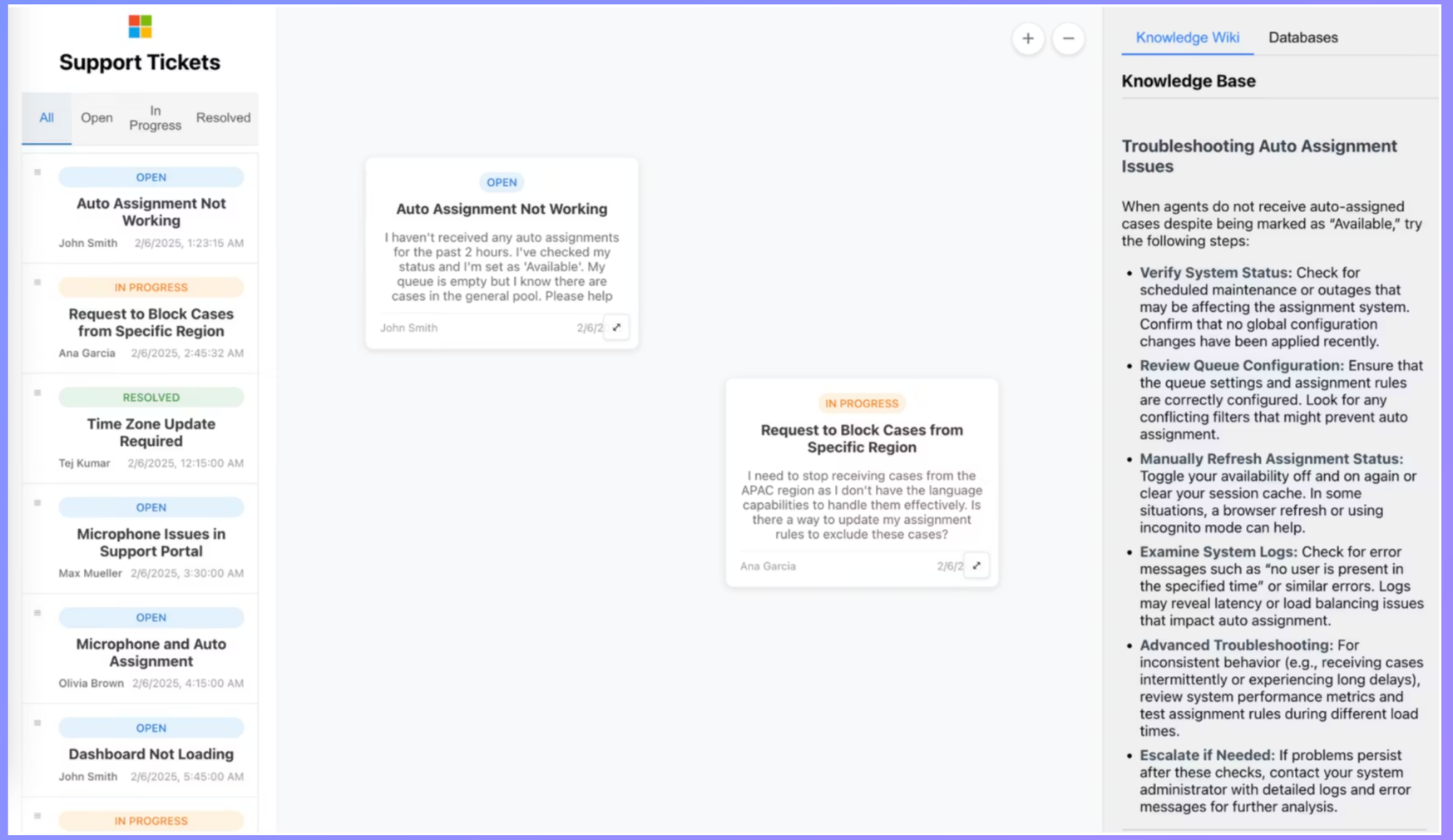
04

IMPACT

AGENTS CAN NOW ANALYZE TICKETS IN ONE PLACE

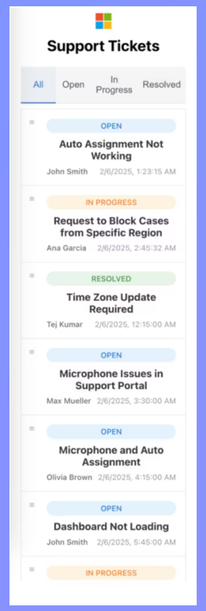
Try Pitch

Solution

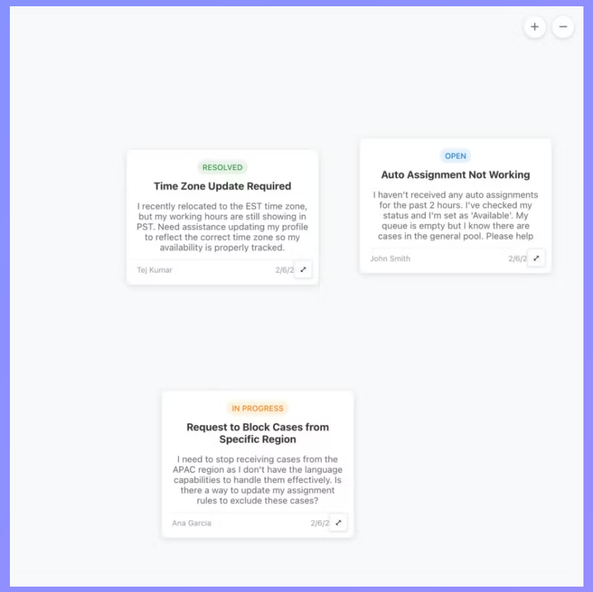


Try Pitch

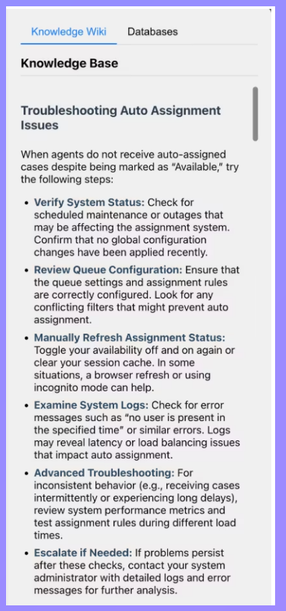
Solution + Key Features



TICKET QUEUE

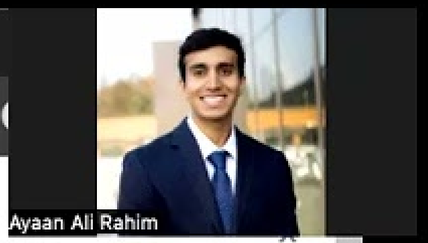


INFINITE TICKET CANVAS



KNOWLEDGE WIKI

Try Pitch



React App

localhost:3000

Support Tickets

All | Open | In Progress | Resolved

OPEN
Auto Assignment Not Working
John Smith 2/6/2025, 1:23:15 AM

IN PROGRESS
Request to Block Cases from Specific Region
Ana Garcia 2/6/2025, 2:45:32 AM

RESOLVED
Time Zone Update Required
Tej Kumar 2/6/2025, 12:15:00 AM

OPEN
Microphone Issues in Support Portal
Max Mueller 2/6/2025, 3:30:00 AM

OPEN
Microphone and Auto Assignment

Try Pitch

OPEN
Auto Assignment Not Working
I haven't received any auto assignments for the past 2 hours. I've checked my status and I'm set as 'Available'. My queue is empty but I know there are cases in the general pool. Please help
John Smith 2/6/2025

administrator with detailed logs and error messages for further analysis. They will be able to assist you in resolving the issue and ensuring that auto-assignment is working correctly for you.

Edit Response

Knowledge Wiki References

- Troubleshooting Auto Assignment Issues

IN PROGRESS
Request to Block Cases from Specific Region
I need to stop receiving cases from the APAC region as I don't have the language capabilities to handle them effectively. Is there a way to update my assignment rules to exclude these cases?
Ana Garcia 2/6/2025

Chat Assistant

confirm any recent updates.
Step 2: Review Queue Configuration and Assignment Rules Ensure that the queue settings and assignment rules are correctly configured. Look for any conflicting filters that might prevent auto-assignment in our **Americas** region (your support region). If you need assistance with this step, your **system administrator** can provide guidance on reviewing these configurations.
Step 3: Manually Refresh Assignment Status Toggle your availability off and on again or clear your session cache. You can do this by clicking on the **Availability** dropdown in your profile settings and toggling it to "Away" and then back to "Available." This should help ensure that any caching issues are cleared.
Step 4: Examine System Logs Check for error messages such as "no user is present in the specified time" or similar errors. To access system logs, you can follow this [link](insert link) to the log viewer portal. If you need assistance with navigating these logs, your **system administrator** can provide support.
Next Steps: If you've completed these steps and are still experiencing issues, please reach out to your **system administrator** with detailed logs and error messages for further analysis. They will be able to assist you in resolving the issue and ensuring that auto-assignment is working correctly for you.

Type your message...

Send

Ethical Considerations

01

VALUES

ACCURACY.
TRUST.
PRODUCTIVITY.

02

POTENTIAL ETHICAL CONCERNS

DATA COMPLIANCE + SENSITIVITY

TRUST LLM-GENERATED INFORMATION

REPLACING HUMAN JOBS

03

WHAT WE DID

PROTECT SENSITIVE USER & COMPANY DATA IN SUPPORT TICKETS,

PROVIDE KNOWLEDGE WIKI CITATIONS,

AI ASSISTS, BUT FINAL DECISIONS SHOULD REMAIN WITH HUMAN AGENTS

SUPPORT AUTOMATION SHOULD ENHANCE, NOT REPLACE HUMAN ROLES

Try Pitch

Next Steps

01

LLM models run local for data security and scale

02

No cloud infrastructure was deployed. No data transfer required.

03

Add project to Capstone archives

Try Pitch

THANK YOU

Try Pitch